

MarketPulse: RAG Pipeline for Stock Price Analysis

Z2004: Database Management Systems | IIT Madras Zanzibar

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Milestone 1: Schema Design, Justification, and Normalisation

Entity Relationship Diagram Justification

Keys and Cardinalities

Our conceptual design models the financial market using precise cardinality and participation constraints:

- **Sectors (1) to Companies (N):** A single sector classifies multiple companies, but a company belongs to exactly one specific sector/industry pairing.
- **Companies (1) to Stock_Prices (N):** A company has many daily stock price records. *Total Participation* is enforced on the *Stock_Prices* side (double line), as a price record cannot exist independently of a company.
- **Companies (1) to Analyst_Ratings (N):** A company receives multiple ratings over time from various analyst firms.
- **Companies (M) to News_Articles (N):** A news article can mention multiple companies, and a company can be mentioned in multiple articles. This many-to-many relationship is resolved in the logical design via the *article_company* junction table.

Handling of Standalone Entities

The *Economic_Indicators* entity handles macroeconomic data sourced from the FRED API. It intentionally lacks a relationship to *Companies*. This is because metrics like the Federal Funds Rate or US GDP apply broadly to the market and are used as standalone semantic context in our Retrieval-Augmented Generation (RAG) pipeline, rather than being mapped to a specific ticker.

Normalisation Argument (3NF)

All tables in the *MarketPulse* database have been demonstrably normalised to the Third Normal Form (3NF).

Criteria Fulfillment

1. **First Normal Form (1NF):** All attributes are atomic. There are no repeating groups or arrays. For example, rather than storing a list of companies mentioned in an article, we utilize an intersection table.
2. **Second Normal Form (2NF):** All tables are in 1NF, and every non-prime attribute is fully functionally dependent on the entire primary key. This is particularly relevant for our *article_company* junction table, which utilizes a composite primary key (*article_id*, *company_id*) with no partial dependencies.

3. **Third Normal Form (3NF):** All tables are in 2NF, and there are no transitive dependencies between non-prime attributes.

Functional Dependencies and Decomposition

To achieve 3NF, transitive dependencies discovered during the initial design phase were eliminated through table decomposition.

Decomposition of Sectors: Initially, `sector_name` and `industry_name` were considered attributes of the `Companies` table. This created a transitive dependency:

$$\text{company_id} \rightarrow \text{sector_id} \rightarrow \{\text{sector_name}, \text{industry_name}\}$$

To eliminate this and satisfy 3NF, we decomposed this into a separate `Sectors` table, linked via a Foreign Key (`sector_id`) in the `Companies` table.

List of Functional Dependencies: The final schema relies on the following functional dependencies, where the left side is the Primary Key (or natural candidate key):

- **SECTORS:** `sector_id` $\rightarrow$ `sector_name`, `industry_name`
- **COMPANIES:** `company_id` $\rightarrow$ `ticker`, `name`, `sector_id`, `exchange`, `market_cap_usd`
(Note: *ticker acts as an alternate/candidate key enforcing uniqueness*).
- **STOCK_PRICES:** `price_id` $\rightarrow$ `company_id`, `trade_date`, `open_price`, `close_price`, etc.
(Note: *{company_id, trade_date} forms a natural candidate key*).
- **NEWS_ARTICLES:** `article_id` $\rightarrow$ `title`, `content`, `url`, `published_at`, `sentiment_score`, `embedding`
- **ANALYST_RATINGS:** `rating_id` $\rightarrow$ `company_id`, `analyst_firm`, `rating`, `rating_date`
- **ECONOMIC_INDICATORS:** `indicator_id` $\rightarrow$ `indicator_name`, `value`, `recorded_date`, `source`, `unit`

By enforcing these dependencies, our schema eliminates update anomalies and guarantees structural integrity for the RAG pipeline.